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Make A Simple Capacitive Touch Keyboard With Arduino

Certain shortcuts on your keyboard such as copy, paste or print can definitely simplify your work. And what if you can create these different keyboard operations on your own? Have you ever thought about this?

This simple DIY project explains how you can make a portable device that performs different keyboard operations like copy and space bar. It is a small single keypad device that works on Arduino and capacitive touch sensor. The sensor can be made using any metallic object like aluminium foil, coin, etc. The author's prototype wired on a breadboard is shown in Fig. 1.

Hardware components include Arduino Micro, 1-mega-ohm resistor, jumper wires, and aluminium foil.

Arduino Micro is the smallest board of the family of ATmega32U4 Arduino. It is easy to integrate with everyday objects to make them interactive. It is based on the ATmega32U4 microcontroller (MCU) featuring a built-in USB, which makes the Micro recognisable as a mouse or keyboard. Note that you can use only the Arduino boards that use ATmega32U4 such as Micro, Leonardo, Leonardo ETH, etc.

Circuit and working

The circuit diagram of the capacitive touch keyboard using Arduino is shown in Fig. 2. Digital pins 2 and 4 of Arduino are used to interface with the sensor.

The sensor setup includes a medium to high-value resistor, some jumper wires, a small piece of aluminium foil. You can use any resistor value between 100-kilo-ohm and 50-mega-ohm. Note that at higher resistor value, it behaves like a proximity sensor.

The circuit requires a capacitive

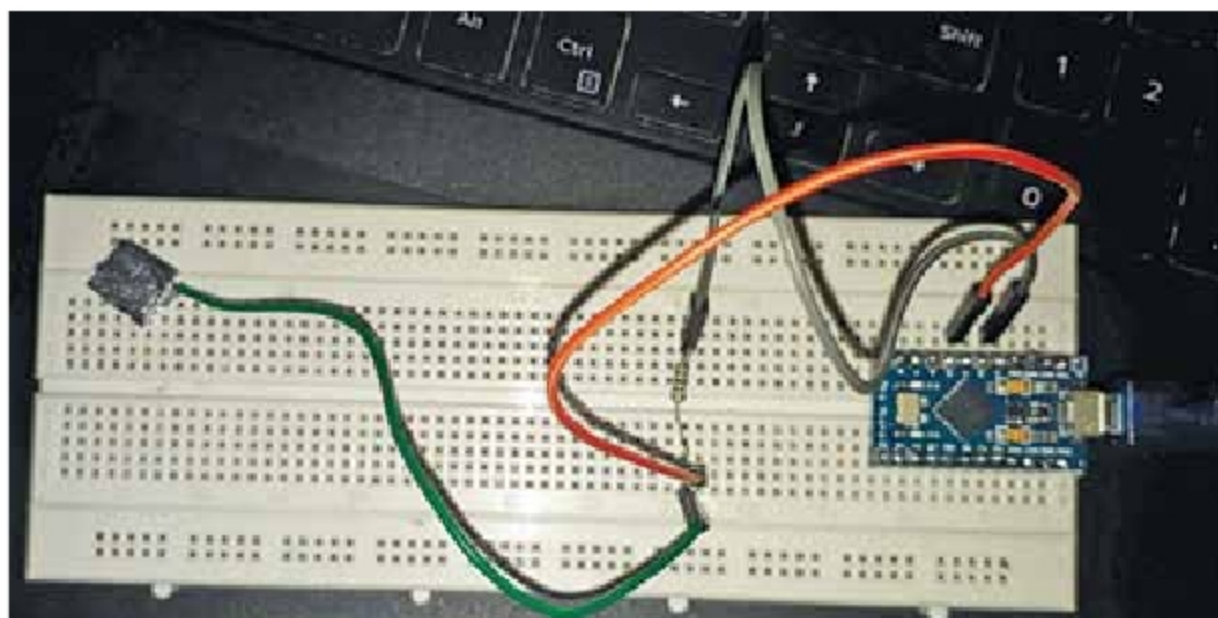


Fig. 1: Author's prototype wired on a breadboard

sensor library in Arduino IDE. The library turns two or more Arduino pins into a capacitive sensor, which can sense the electrical capacitance of the human body. Capacitive sensor can detect anything that is conductive or that has a significantly different permittivity than air, such as the human body or hand.

The project was tested using a 1-mega-ohm resistor. If you are using a different resistor value, use the program (capturing_sensor_value.ino) to find the sensor value before uploading the main Arduino sketch to Arduino board.

Software program

Software is written in Arduino programming language. The capturing_sensor_value.ino code is used to decode the sensor value. The copy_operation.ino code is used for copy operation and the space_bar.ino code is used for space bar operation on the keyboard.

After installing Arduino IDE in your PC, launch Arduino IDE Software, connect Arduino Micro to your PC, select the board (Arduino/Genuino Micro) and COM port. Then install capacitive sensor library, which

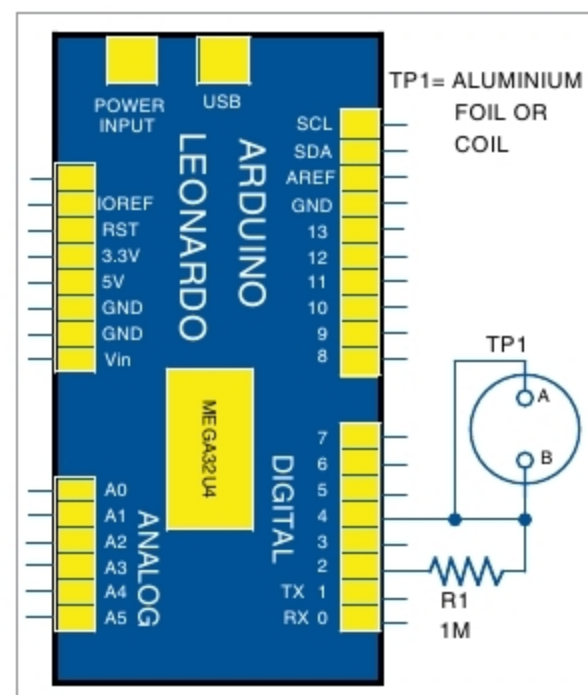


Fig. 2: Circuit diagram of the capacitive touch keyboard

can be downloaded from the link <https://github.com/arduino-libraries/CapacitiveSensor/zipball/master>

For obtaining the sensor values, open capturing_sensor_value.ino from Arduino IDE, compile and upload to Arduino board with the sensor connected to the board. Next, open serial monitor of Arduino IDE and observe the values. The default sensor values obtained on the serial monitor without touching the sensor